

AUTOMATED IDENTIFICATION OF GAIT ABNORMALITIES

UE24CS352A — Machine Learning Mini-Project Report

GitHub Repository

<https://github.com/ChigilarNishanth/ML-MiniProject-GaitAnalysis>

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Abstract

Gait abnormalities are key clinical indicators for progressive neuromuscular pathologies such as Cerebral Palsy, Parkinson's disease, and stroke. The Gait Deviation Index (GDI) quantifies gait pathology on a continuous scale from 0 to 100. Traditional clinical GDI measurement relies on expensive, marker-based 3D motion capture systems in specialized laboratories. This project implements **GDI-Net**, an automated, end-to-end spatiotemporal deep learning framework that estimates GDI directly from monocular RGB video captured on commodity devices. By featurizing video frames using DensePose pose representations (IUV coordinates) and coupling a custom 2D Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network, the system achieves a validation Root Mean Squared Error (RMSE) of 4.4 and a coefficient of determination (R^2) of 0.86.

1. Problem Statement & Motivation

Neuromuscular pathologies significantly impair human locomotion, resulting in observable irregularities in step cadence, stride length, joint angular velocities, and postural stability. Biomechanists and orthopedic surgeons evaluate disease progression using the Gait Deviation Index (GDI), where a score of 100 represents a non-pathological gait and each 10-point decrement indicates one standard deviation from normal gait kinematics.

The current gold standard for assessing GDI requires placing physical reflective markers on patient anatomical landmarks and recording movement via multi-camera array systems. This process presents major clinical hurdles:

- **High Equipment Costs:** Motion capture laboratories require specialized multi-camera rigs and force plates costing hundreds of thousands of dollars.
- **Time and Physical Burden:** Assessment sessions require several hours of setup and calibration, creating discomfort for pediatric or severely impaired patients.

Project Objective: Develop a low-cost, automated machine learning pipeline to predict continuous GDI scores (0–100) directly from single-camera monocular video clips, providing physicians with a rapid screening tool.

2. Dataset & DensePose Preprocessing

Dataset Specifications

The model was trained and evaluated using clinical video data collected at Gillette Children’s Specialty Healthcare Center for Gait and Motion Analysis:

- **Data Volume:** Approximately 3,249 monocular RGB video clips recorded at a resolution of 640×480 pixels at 25 frames per second (FPS).
- **Ground Truth Labels:** Continuous scalar GDI scores (0–100) derived from full clinical marker-based assessments conducted by medical professionals.
- **Data Partitioning:** Standard 80% training (2,599 videos) and 20% validation (650 videos) split.

DensePose Feature Extraction

Standard RGB camera feeds contain background clutter, clothing variability, and lighting noise that degrade regression performance. To isolate anatomical movement, every frame was processed using DensePose-RCNN.

Raw RGB Frame ($640 \times 480 \times 3$) → DensePose-RCNN (SMPL Model) → IUUV Tensor ($480 \times 640 \times 3$)

DensePose maps pixels from the 2D image plane to the 3D surface parameterization of the human body using the Skinned Multi-Person Linear (SMPL) model. The output is a 3-channel frame representation:

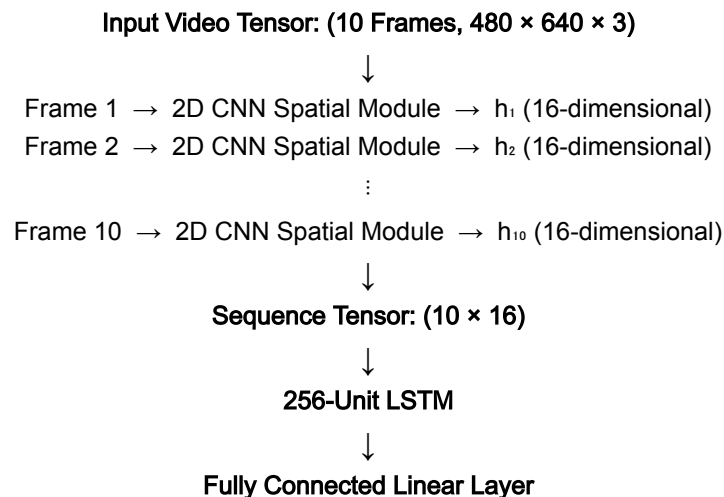
- **Channel I (Part Index):** Identifies which of the 24 segmented body parts a pixel belongs to.
- **Channels U & V (Texture Coordinates):** Local 2D surface parameterization coordinates mapping exact locations on the 3D human mesh.

Sequence Construction

To model temporal dynamics without incurring memory limits, videos were processed into 10-frame consecutive tensor clips of dimensions (10, 480, 640, 3).

3. Methodology & Model Architecture (GDI-Net)

Pathological gait analysis requires extracting both individual frame spatial configurations (e.g., knee flexion, hip angle) and inter-frame temporal kinematics (e.g., cadence, foot-strike velocity). **GDI-Net** integrates a 2D CNN spatial featurizer with a 256-unit Recurrent LSTM sequence model.



↓
Predicted GDI Score

Spatial Component (Custom 2D CNN)

Each of the 10 DensePose frames is individually fed through a weight-shared custom 2D CNN to compress high-dimensional spatial IUV features:

1. **Block 1:** $2 \times \text{Conv2D}$ (kernel = 3×3 , filters = 8) $\rightarrow$ ReLU $\rightarrow$ Batch Normalization $\rightarrow$ MaxPool2D (2×2) $\rightarrow$ Dropout (0.7).
2. **Block 2:** $2 \times \text{Conv2D}$ (kernel = 3×3 , filters = 8) $\rightarrow$ ReLU $\rightarrow$ Batch Normalization $\rightarrow$ MaxPool2D (2×2) $\rightarrow$ Dropout (0.7).
3. **Block 3:** $2 \times \text{Conv2D}$ (kernel = 3×3 , filters = 8) $\rightarrow$ ReLU $\rightarrow$ Batch Normalization $\rightarrow$ MaxPool2D (4×4) $\rightarrow$ Dropout (0.7).
4. **Compression:** Flatten $\rightarrow$ 16,960 features $\rightarrow$ Fully Connected Dense (ReLU activation) $\rightarrow$ 16-dimensional spatial vector per frame.

Temporal Component (LSTM)

The 10 individual 16-dimensional vectors are concatenated in chronological order into a (10, 16) sequence array. A 256-unit Recurrent LSTM network processes this sequence, tracking trajectory dynamics and angular displacement over time.

Output Layer

The final hidden state of the LSTM feeds into a single fully connected neuron with a linear activation function to predict the continuous GDI scalar value:

$$\hat{y} = \mathbf{W}^T \mathbf{h}_{\text{LSTM}} + b$$

4. Implementation & Training Details

- **Software Framework:** Implemented in Python using Keras with a TensorFlow backend, and Scikit-Learn for evaluation metrics.

Loss Function: Trained to minimize Mean Squared Error (MSE) between ground truth y_i and predicted $\hat{y}_i$:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Optimization & Learning Rate: Optimized using the Adam optimizer with an initial learning rate $\eta_0 = 0.01$, decaying linearly to $\eta = 0.001$ across 100 epochs.

- **Regularization:** Heavy dropout (0.7) applied after every spatial pooling block to prevent overfitting on specific patient background artifacts. Batch size was constrained to 4 due to GPU memory limitations during sequence loading.

5. Experimental Results & Discussion

Model performance was evaluated using Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Model Performance Comparison

Model Architecture	Input Sequence Type	Input Dimensions	Training RMSE	Validation RMSE
Guess Mean Baseline	Single Frame	(480 × 640 × 3)	13.7	13.7
Linear Regression	5 Frames Concatenated	Vector (4,608,000)	0.0	13.0
VGG16 (Pre-trained Transfer Learning)	Single Cropped Frame	(224 × 224 × 3)	10.1	9.5
VGG16 (Re-trained Weights)	Single Cropped Frame	(224 × 224 × 3)	11.6	11.4
Custom 2D CNN	Single Full Frame	(480 × 640 × 3)	8.1	8.2
Custom 2D CNN + 1D CNN	10-Frame Video Clip	(10 × 480 × 640 × 3)	4.2	11.3
GDI-Net (Custom 2D CNN + LSTM)	10-Frame Video Clip	(10 × 480 × 640 × 3)	1.4	4.4

Key Experimental Insights

- 1. Failure of Transfer Learning (VGG16):** Off-the-shelf VGG16 models performed poorly (Validation RMSE = 9.5–11.4). Pre-trained ImageNet weights are optimized for standard RGB color textures rather than DensePose IUUV coordinate spaces. Additionally, cropping frames to 224 × 224 removed peripheral limb position data.
- 2. Necessity of Temporal Recurrence:** Single-frame spatial models (Custom 2D CNN) stagnated at an RMSE of 8.2. Because GDI depends on temporal gait phase changes (stance phase vs. swing phase), temporal tracking is mandatory.
- 3. Superiority of GDI-Net:** Combining spatial CNN featurization with an LSTM temporal sequence model reduced validation error to 4.4 with an $R^2 = 0.86$, closely capturing ground-truth clinical GDI variance.

6. Conclusions & Future Scope

This project demonstrates that deep spatiotemporal models operating on monocular DensePose video sequences can accurately estimate Gait Deviation Index scores without expensive motion capture equipment.

Future Work

- 1. 3D Convolutional Networks (C3D):** Replacing separate 2D CNN and LSTM blocks with unified 3D convolutional kernels to extract joint spatiotemporal features simultaneously.
- 2. Kinematic Parameter Extraction:** Converting DensePose UV arrays to explicit 3D joint coordinate vectors (X, Y, Z) to explicitly compute physiological parameters like knee flexion angle and hip rotation.
- 3. Probabilistic Classification:** Bucketing GDI scores into discrete integer classes and computing the expected GDI scalar using a probability-weighted softmax average.